

GPT and Autoregressive LM

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NLP

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Autoregressive Language Models

- **Autoregressive LM**: Predict the next token based only on previous tokens (causal modeling).
- **Objective (Causal Language Modeling)** : predict next token at each position.

  Maximize

$$\sum_i \log(p(w^i | w^1, \dots, w^{i-1})$$

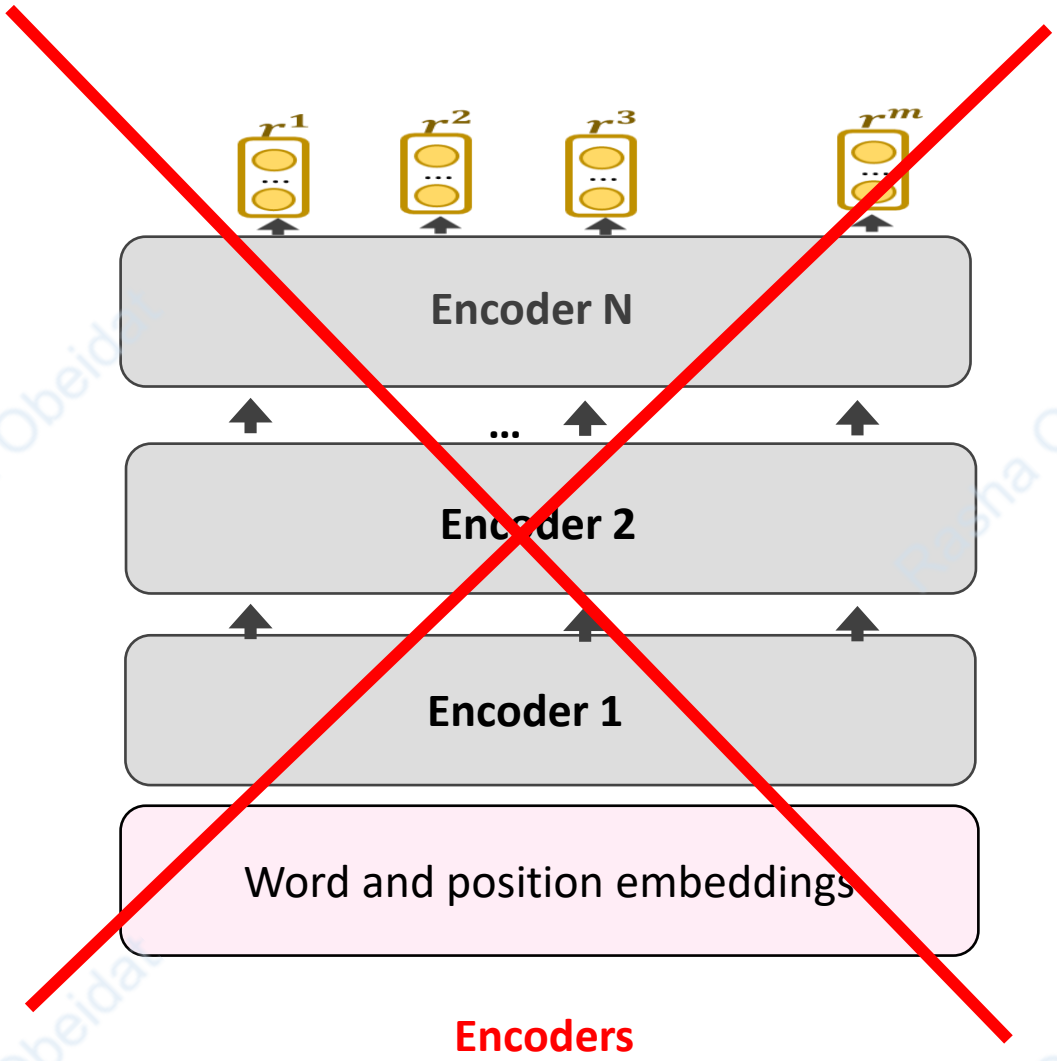
- No access to future tokens.
- Examples:
  - **GPT-1** to **GPT-4**
  - **Reformer**: efficient Transformer variant that enables processing very long sequences.
  - **CTRL**: controlled text generation via conditions.
  - **XLNet**: combines autoregression with permutation for better bidirectional context.

Transformers: Encoder vs Decoder

- Transformer **Encoder**:
  - Bidirectional attention (sees all tokens).
- Transformer **Decoder**:
  - Masked attention (causal, future is hidden).
- Decoder used for generation.

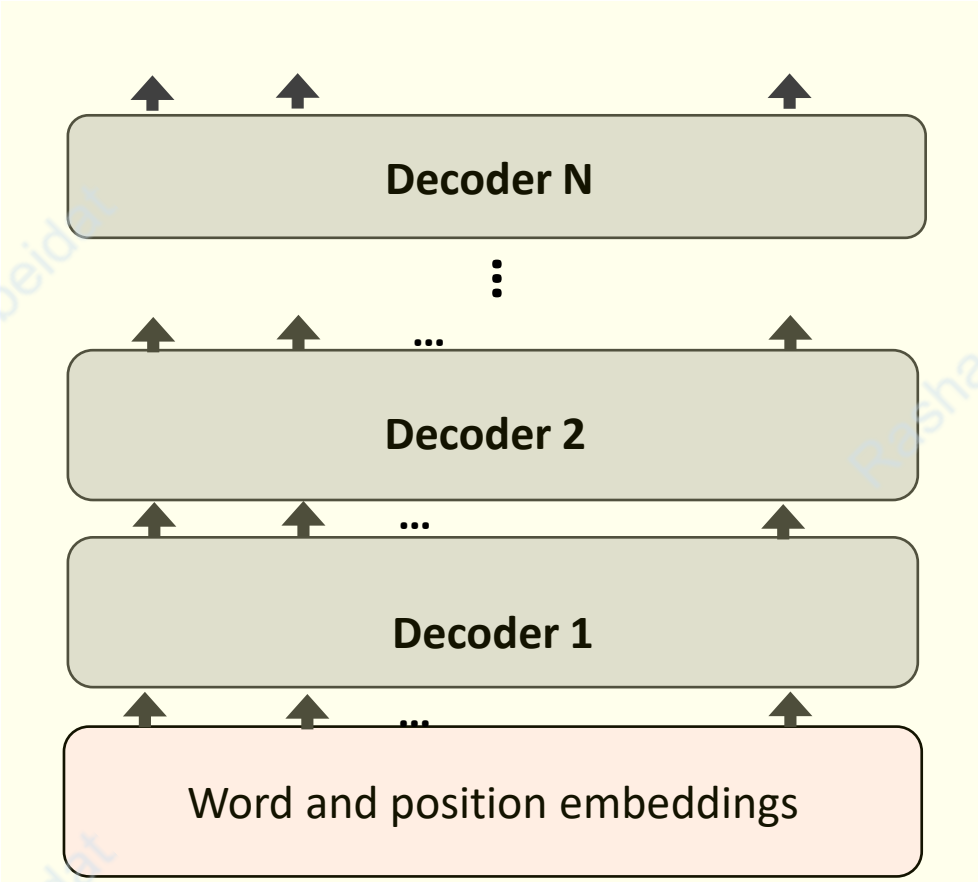
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GPT = Transformer Decoder Only



Encoders

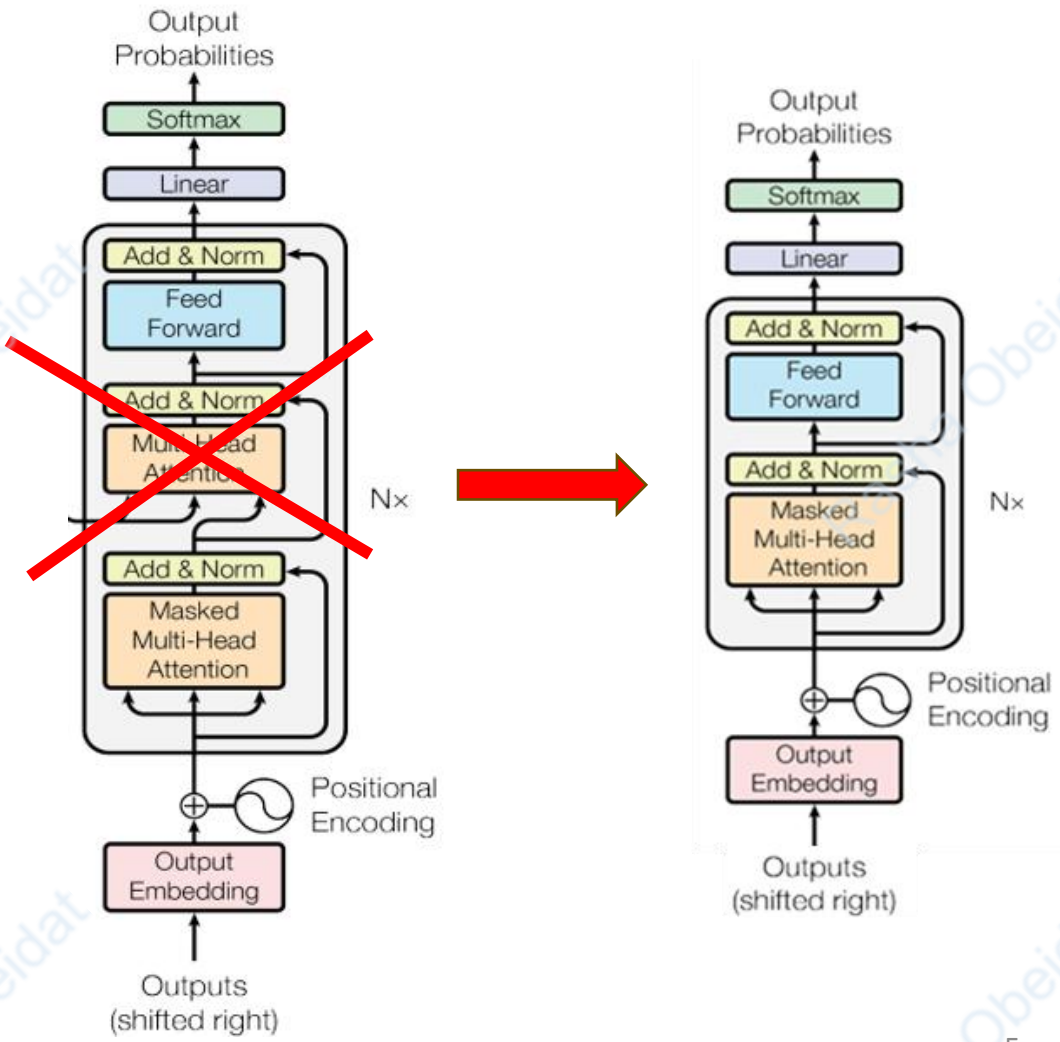
GPT only uses the decoder



Decoders

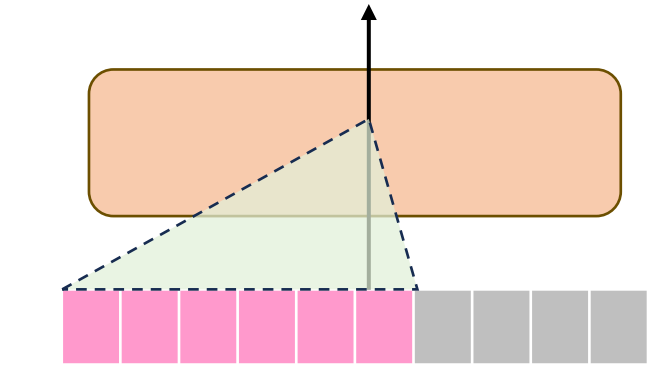
GPT = Transformer Decoder Only

- GPTs uses **only** the decoder stack.
- **Cross-attention is removed**, not replaced.

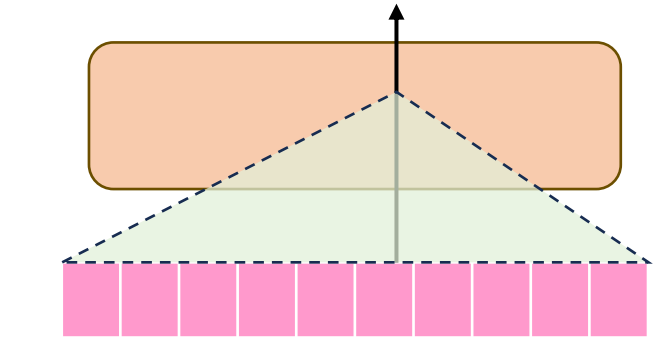


GPT Uses Masked Self-attention

- Unlike BERT, which uses bidirectional self-attention that attend to all tokens in the input sequence.
- GPTs uses masked self-attention, to predict the next token where each token is only allowed to attend to **itself and previous tokens**. This ensures that GPT cannot access future tokens during training or generation.



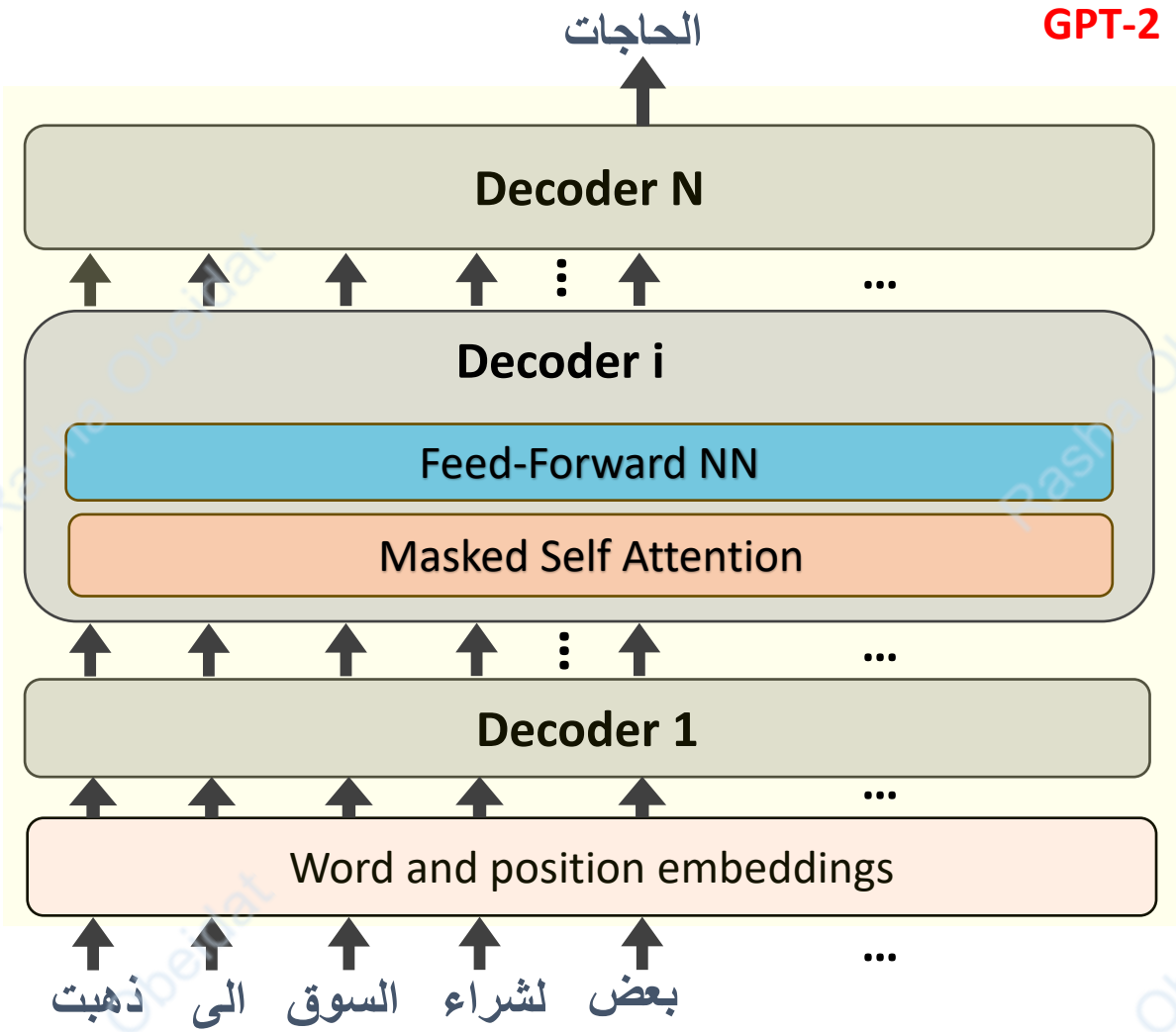
Masked Self-attention in GPT-2



Bidirectional Self-attention in BERT

GPT Predicts the Next Token

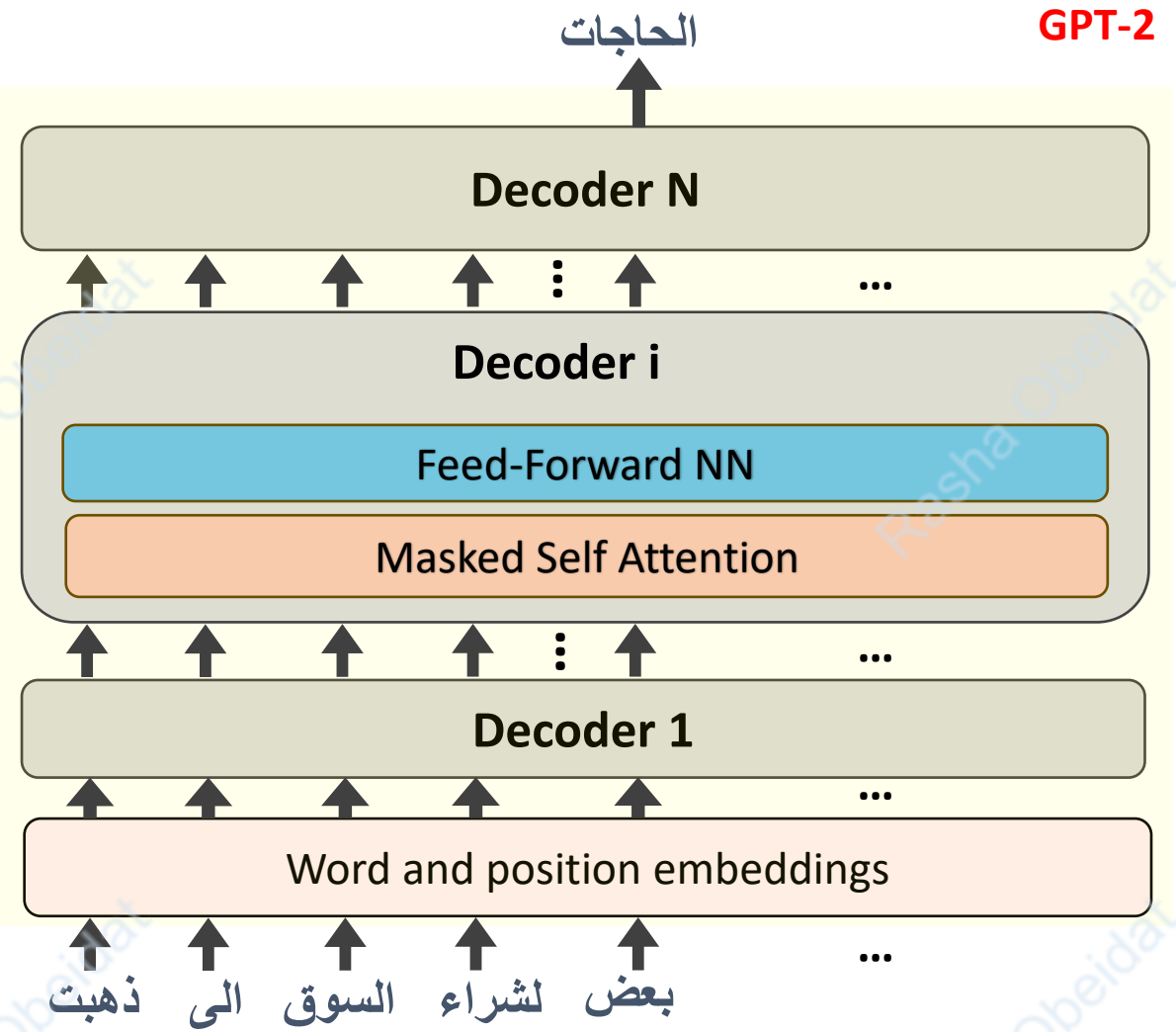
- Unlike BERT trained to predict **current masked token** (in MLM),
- GPTs is trained predict the **next token** given the current and previous tokens .



GPT-2 Predicts the Next Token

- **BERT** has **N encoder blocks**, each containing **multi-head self-attention**. This mechanism combines **12 parallel single attention heads**, each producing a representation of each token with a dimensionality of **64**. These outputs are concatenated, resulting in a **final token** representation of **768 dimensions**.

- **GPT-2** (for example) has **N decoder blocks**, each using **masked multi-head self-attention**, also composed of **12 masked single attention heads**, each with a dimensionality of **64**. The outputs are similarly concatenated, yielding a **768-dimensional** representation per token.



Input Representation in GPT-(1-3)

1. Token Embeddings

- Input text is tokenized using **Byte Pair Encoding (BPE)**
  (e.g., "playing" → "play", "ing")
- **BPE** Looks at the individual symbols (e.g., characters) and use agglomerative clustering to repeatedly merges the most frequent pairs.
- Each subword token (not full words or characters) is mapped to a learned embedding
- This is conceptually similar to **WordPiece** in BERT, used to handle rare words (OOV).
- Unlike WordPiece, **BPE** does not use special markers (e.g., **"##"**) to indicate continuation.

2. Positional Embeddings

- Since GPT is **autoregressive**, it relies heavily on positional embeddings.
- Each token is assigned a unique position index (e.g., 0, 1, 2, ...).

- **Note:** this mostly apply on GPT-3.5 and GPT-4 (though some internal details of GPT-3.5 and GPT-4 are not public).

Training GPTs

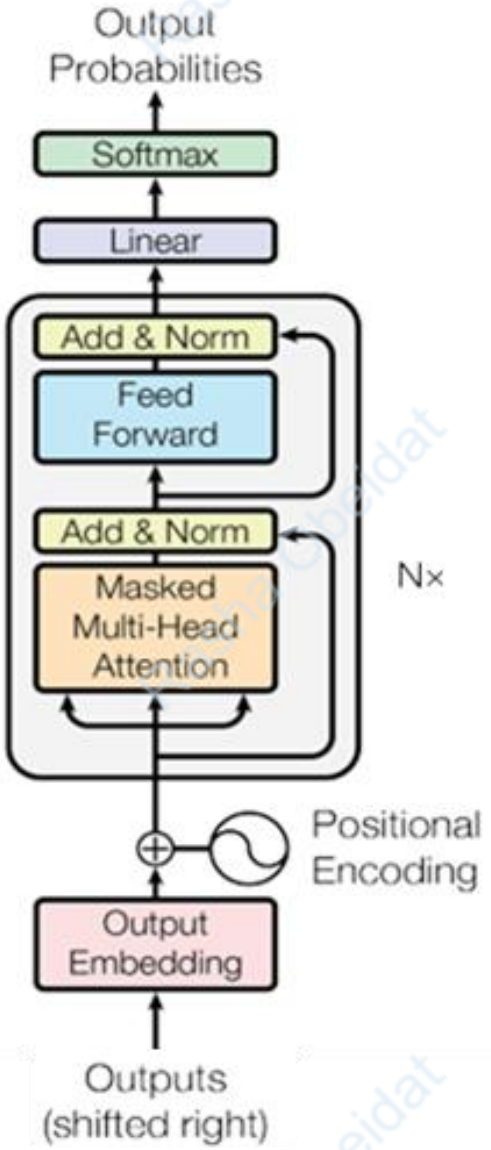
- Autoregressive (Causal) Language Modeling Objective:

$$\sum_i \log(p(x^i | x^1, \ldots, x^{i-1})$$

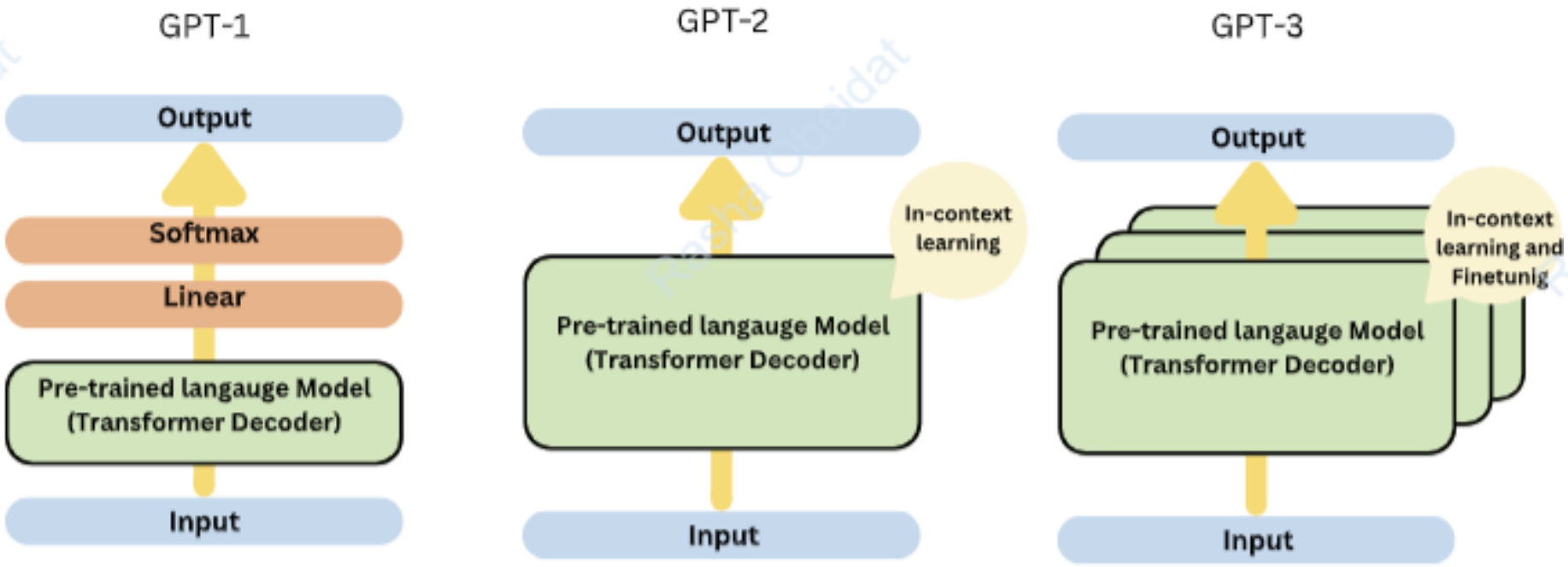
  - Predicts the **next token** based only on **previous tokens.**
  - Is applied during pretraining across all versions: GPT-1, GPT-2, GPT-3, GPT-3.5, GPT-4.

GPT Structure and Scaling Design

- **Pre-Layer** Normalization**.**
  - GPT-1 uses normal (**Post-LayerNorm)** that is the post-layer normalization originally used in Transformer.
  - GPT-2, GPT-3 and GPT-3.5 switched to **Pre-LayerNorm instead** of Post-LayerNorm (Undisclosed for GPT-4). I.e, it applies **Layer Normalization before** the attention and feedforward sublayers inside each decoder block which improves stability during deep training.
- **Deep stacking** and **wide** hidden states
  - As we move from GPT-2 to larger GPT-3 and then to GPT-4, the **number of layers** and the **width of hidden states** increase significantly.
  - Example: GPT-2 (12–48 layers, 768–1600 hidden size) → GPT-3 (up to 96 layers, 12288 hidden size).
  - This architecture allows **smooth scaling** of model size and performance.



Evolution of the GPT Family



Bansal, Gaurang, et al. "Transforming Conversations with AI—A Comprehensive Study of ChatGPT." *Cognitive Computation* (2024).

GPT-1

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GPT-1

- **Model**:
  - Transformer decoder
  - 117M Parameters
  - 12 Layer
  - Max sequence length: 512
  - Uses **Post-LayerNorm** → LayerNorm is applied **after** attention and feedforward blocks (original Transformer architecture).

Improving Language Understanding by Generative Pre-Training

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- **Release:** 2018
- **Data**: **BooksCorpus** which is a collection of ~7,000 books across various genres such as adventure, fantasy, and romance (~ 5– 6 GB).
- **Focus**:
  - Improving ***Language Understanding*** by ***Generative Pre-Training***.
  - fine-tuning the pre-trained GPT-1 on supervised tasks to achieve SOTA results.

GPT-1 Discriminative Fine-tuning

- GPT-1 demonstrated that unsupervised pretraining could produce useful representations ***transferable*** to many NLP tasks.
- GPT-1 Made a strong case for transfer learning in NLP, similar to **ImageNet** in vision.
- **Fine-Tuning for Downstream Tasks**: after pretraining, GPT-1 was fine-tuned separately on each supervised NLP task:
  - Classification (e.g., sentiment analysis)
  - Natural Language Inference
  - Semantic Similarity
  - Multiple choice QA

GPT-1 Discriminative Fine-tuning

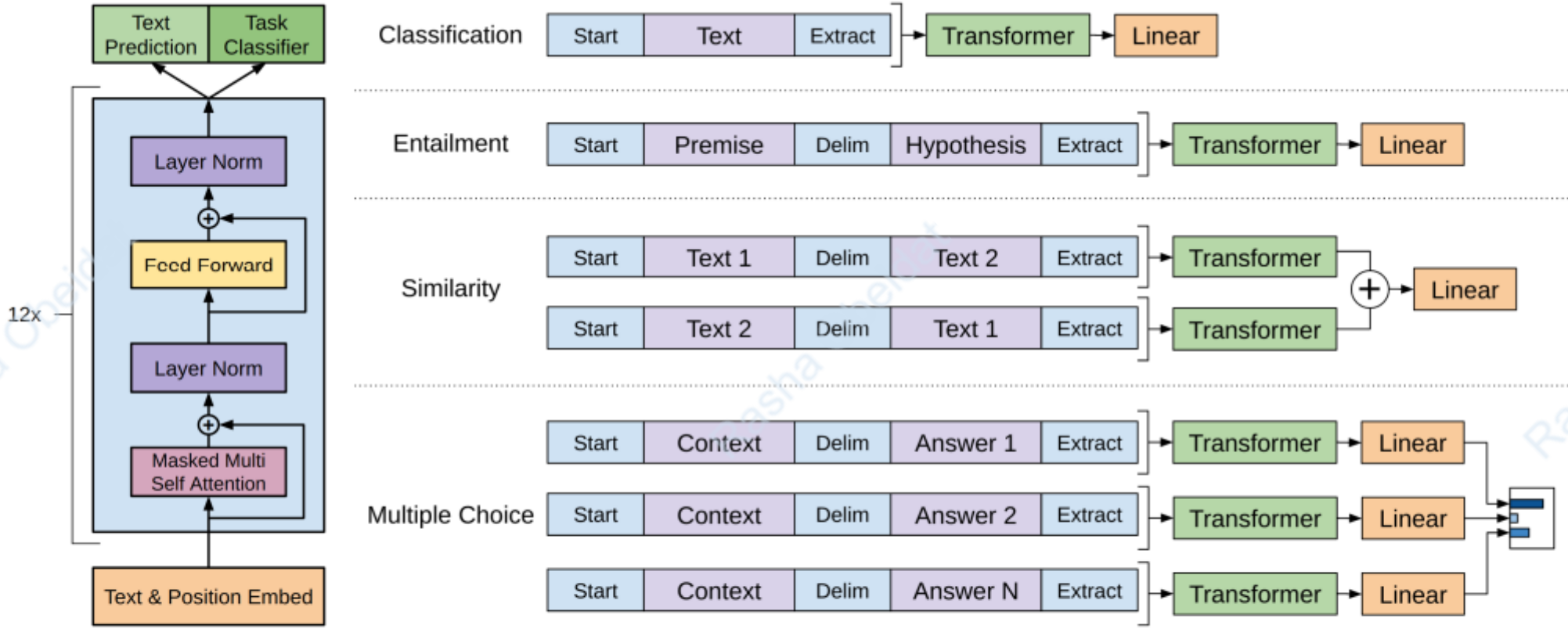
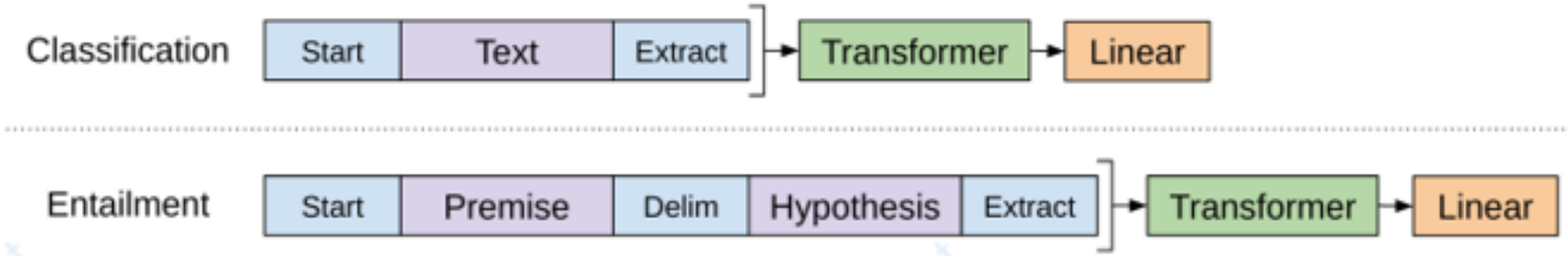


Figure 1: **(left)** Transformer architecture and training objectives used in this work. **(right)** Input transformations for fine-tuning on different tasks. We convert all structured inputs into token sequences to be processed by our pre-trained model, followed by a linear+softmax layer.

https://cdn.openai.com/research-covers/language-unsupervised/language_understanding_paper.pdf

GPT-1 Discriminative Fine-tuning

- **Input Format**
  - special tokens are used to specify start of input (**Start**), Boundary between segments (**$** is used to separate premise from hypothesis in NLI) and indicate where model predictions should begin during supervised fine-tuning (**Extract**).
- **Classification Tasks (SA and NLI)**
  - Task-specific output layer (with Softmax) is added on top of the ***final hidden state*** of the last token (extract) to generated probability distribution over the task-specific labels.



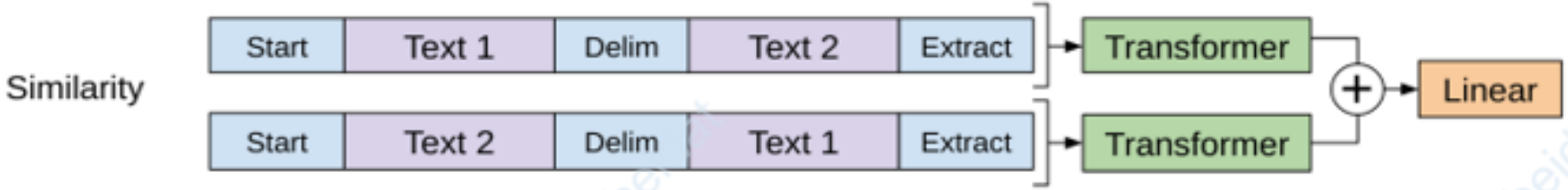
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GPT-1 Discriminative Fine-tuning

- **Semantic Similarity task**
  - **Goal:** Predict a similarity score between two sentences.
  - **Input**: Two input sequences are created with reversed sentence orderings:
  - This avoids introducing bias from sentence order, since similarity is **symmetric**.
  - The model produces a hidden state h^m for the <extract> token (i.e., the final token in the sequence).
  - **Aggregation:** The two hidden vectors from the two input orderings are added element-wise:

$$h = h^{Text1 \rightarrow Text2} + h^{Text2 \rightarrow Text1}$$

  - **Prediction:** the combined vector h is passed to a linear output layer to predict a similarity score. Typically trained with regression loss (e.g., mean squared error or cross-entropy over rounded labels).



https://cdn.openai.com/research-covers/language-unsupervised/language_understanding_paper.pdf

GPT-1 Discriminative Fine-tuning

Sentiment Analysis (CoLA, SST2)

Semantic similarity

NLI

Method	Classification		Semantic Similarity			GLUE
	CoLA (mc)	SST2 (acc)	MRPC (F1)	STSB (pc)	QQP (F1)	
Sparse byte mLSTM [16]	-	**93.2**	-	-	-	-
TF-KLD [23]	-	-	**86.0**	-	-	-
ECNU (mixed ensemble) [60]	-	-	-	81.0	-	-
Single-task BiLSTM + ELMo + Attn [64]	35.0	90.2	80.2	55.5	66.1	64.8
Multi-task BiLSTM + ELMo + Attn [64]	18.9	91.6	83.5	72.8	63.3	68.9
Finetuned Transformer LM (ours)	**45.4**	91.3	82.3	**82.0**	**70.3**	**72.8**

Stanford Sentiment Treebank

Method	MNLI-m	MNLI-mm	SNLI	SciTail	QNLI	RTE
ESIM + ELMo [44] (5x)	-	-	89.3	-	-	-
CAFE [58] (5x)	80.2	79.0	89.3	-	-	-
Stochastic Answer Network [35] (3x)	80.6	80.1	-	-	-	-
CAFE [58]	78.7	77.9	88.5	83.3		
GenSen [64]	71.4	71.3	-	-	82.3	59.2
Multi-task BiLSTM + Attn [64]	72.2	72.1	-	-	82.1	**61.7**
Finetuned Transformer LM (ours)	**82.1**	**81.4**	**89.9**	**88.3**	**88.1**	56.0

https://cdn.openai.com/research-covers/language-unsupervised/language_understanding_paper.pdf

GPT-1 Pretraining and Fine-tuning Objectives

- **Unsupervised Pretraining Objective**:
  - Autoregressive (Causal) Language Modeling using context window)
  - **Input**: unsupervised corpus of tokens $x = \{x^1, x^2, \dots, x^n\}$

$$L_1 = \sum_i \log(p(x^i | x^{i-k}, \dots, x^{i-1}; \theta)$$

  Where:

  - k is the size of the context window (can be replaced with varying sentence length).
  - P conditional probability, modeled using a neural network with parameters θ.
- These parameters are trained using stochastic gradient descent.

GPT-1 Pretraining and Fine-tuning Objectives

- **Supervised fine-tuning Objective**:
  - **Input**: a labeled dataset C, where each instance consists of a sequence of input tokens, $x^1, \dots, x^m$, along with a label y

$$P(y|x^1, \dots, x^m) = softmax(h^m W_y)$$

$$L_2 = \sum_i \log(p(y\,|x^1, \dots, x^m; \theta)$$

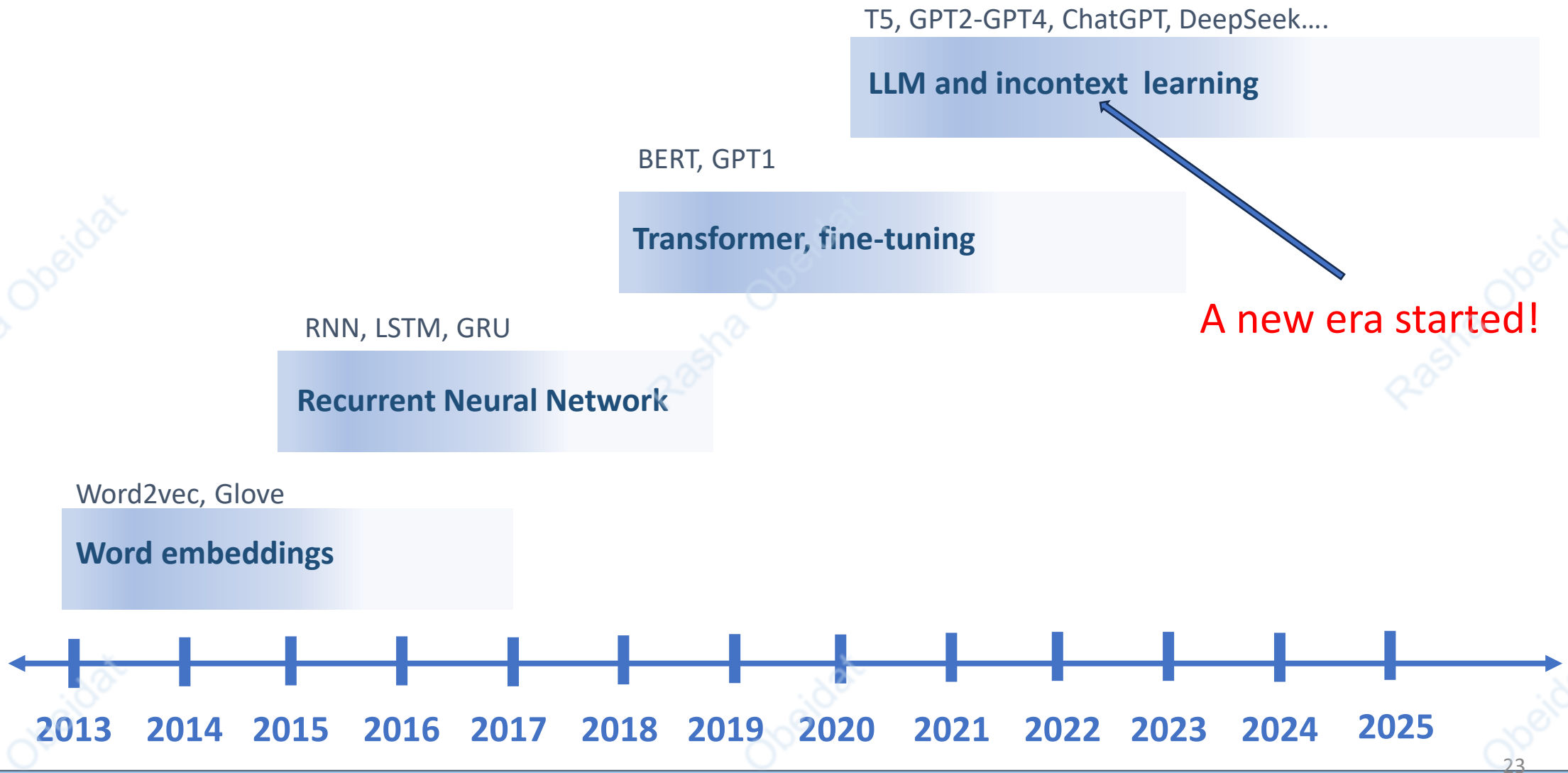
  - It has been *found* that including **language modeling objective ($L1$)** as an **auxiliary objective** to the **fine-tuning** helped learning, by:
    1. Improving generalization of the supervised model.
    2. Accelerating convergence.
  - The **final objective function** optimized for fine-tuning (with weight λ):

$$L_3(C) = L_2(C) + \lambda L_1(C)$$

GPT-2

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Neural Representation Learning for NLP



GPT-2

- **Release:** 2019

Language Models are Unsupervised Multitask Learners

Alec Radford [*1] Jeffrey Wu [*1] Rewon Child [1] David Luan [1] Dario Amodei [**1] Ilya Sutskever [**1]

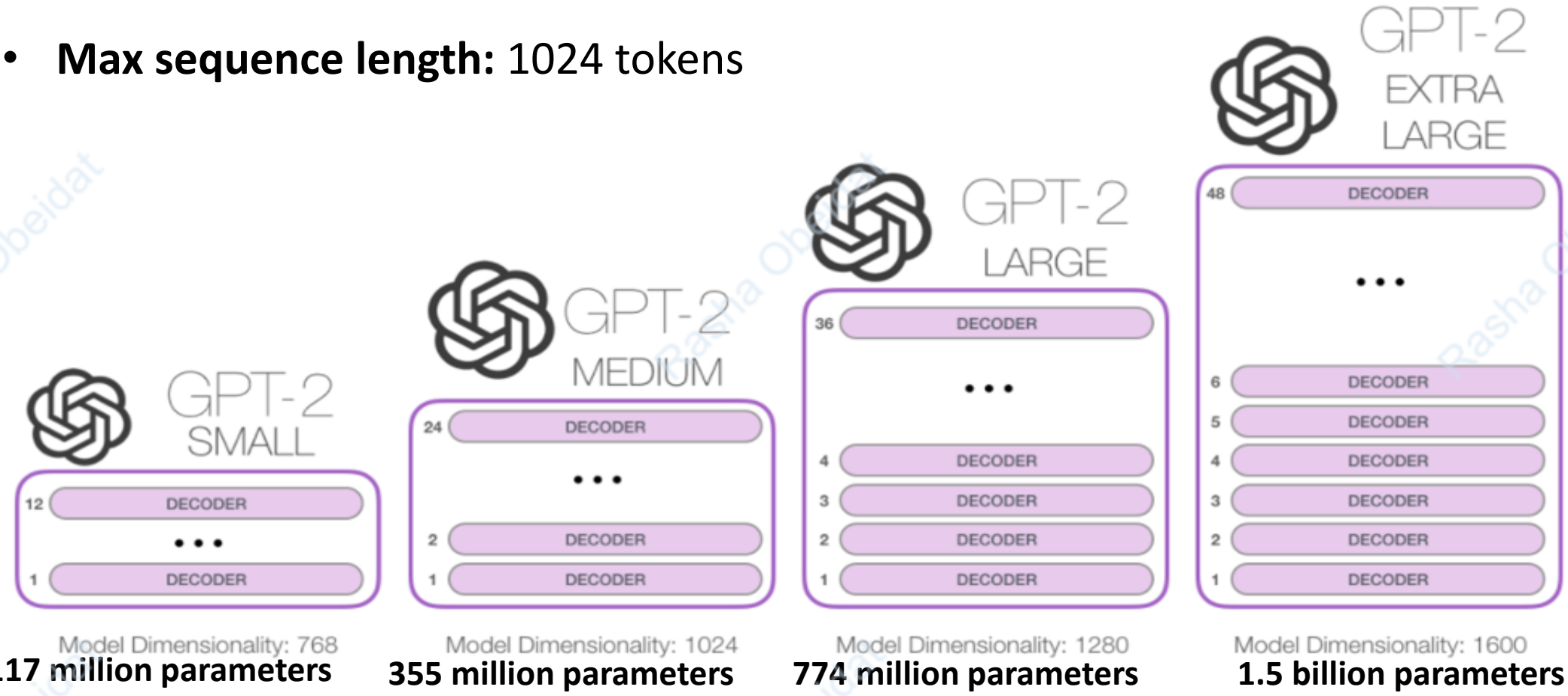
- **Objective**: Same as GPT-1, an Autoregressive (*causal*) Language Modeling objective using cross-entropy loss.
- **Data**: WebText (a curated dataset of 8+ million high-quality web pages, filtered for content richness and diversity).
- **Model**:
  - Transformer decoder and Pre-Layer normalization.
  - Four versions (various depths and parameter counts).

GPT-2 Focus

- Demonstrating that large-scale **Unsupervised Multitask learning** from pretraining alone can enable strong zero-shot performance across NLP task.
  - scaling up model size and training data.
  - no gradient updates.
  - no task-specific fine-tuning .
  - just task-formatted prompts.
- **Unsupervised:** No labeled task-specific data is used during training.
- **Multitask:** GPT-2 is trained only to predict the next token, yet it can perform many tasks (e.g., QA, translation, summarization) at inference time **without fine-tuning**, simply by using the right **prompt**.

Scaling Up GPT-2

- **Max sequence length:** 1024 tokens



The Illustrated GPT-2 : https://jalammar.github.io/illustrated-gpt2/

Key Innovations over GPT-1

Scaled up massively:

- More data

Model	**Source**	**Size**	**Richness & Coverage**
GPT-1	BooksCorpus (fiction)	~ 5– 6GB (~800M words)	Narrow domain (long-form fiction); limited diversity
GPT-2	WebText (Reddit links)	~40GB (~8M docs)	Broad domain (news, Q&A, forums); high linguistic and task diversity

- Larger model sizes (117M → 177M, then 355M, then 744M and 1.5B parameters).
- Longer context (up to 1600 tokens)
- Bigger batch sizes.

Zero-Shot Evaluation

- Evaluate the model on a new task **without any additional task-specific fine-tuning**, using only natural-language task descriptions (**prompts**).
- GPT-2 tested whether **scaling** and **pretraining alone** could produce a model capable of **few-shot and zero-shot** generalization.
- Transferring from NLU (GPT-1) to NLG, which is more challenging!
- Achieved SOTA on most of NLG dataset compared with tuned model.

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Prompting GPT-2: Examples

- Instead of training the model on labeled data, GPT-2 is given **input in a natural task-like prompt**.
- GPT-2 is expected to complete the sequence with the correct output.

Task: Text Classification (Sentiment)

Prompt:	**Review: This movie was terrible.** **Sentiment:**
Model completes	**Negative**

Task: Machine Translation

Prompt:	**Translate English to Arabic** **The book is on the table ⇒ .** **Arabic:**
Model completes	الكتاب على الطاولة.

Prompting GPT-2

Generate the Completion

- You pass the prompt into GPT-2.
- The model predicts the **next tokens** until a stopping condition (e.g., newline, punctuation, max length).
- The generated output is **taken as the answer**.

Prompting GPT-2 (Example)

- **Input Prompt (given to GPT-2):**

  The sun is the star at the center of the Solar System. It is a nearly perfect sphere of hot plasma, and it provides the most important source of energy for life on Earth. The sun's gravity holds the Solar System together, keeping everything from the biggest planets to the smallest particles of debris in its orbit. TL;DR:

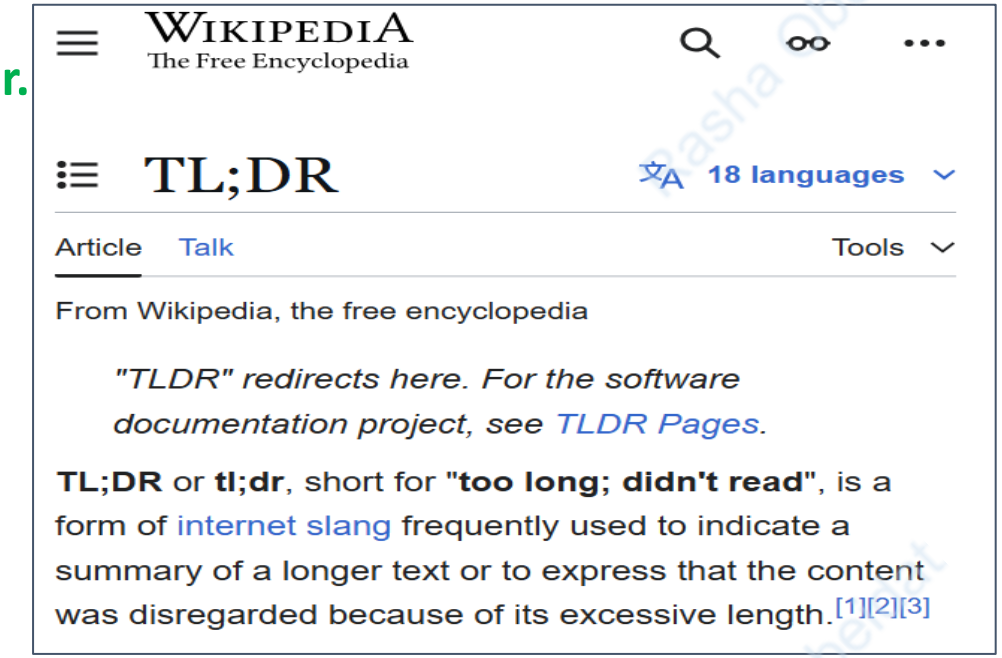
  The colon after TL;DR is important!

- **Model Output (completion):**

  The sun is a hot star that holds the solar system together.

- **Stopping Conditions for Generation**

Max tokens	e.g., stop after 20 or 50 generated tokens
End punctuation	e.g., stop after the first full stop or newline \n
Special prompt	e.g., after TL;DR: + one sentence, stop if next token is newline
Manual truncation	Paper authors may truncate generations to match summary length (e.g., 1-2 sentences)



GPT-2 Zero-shot Results

- Achieved **strong zero-shot performance** on:
  - Question answering
  - Translation
  - Summarization
  - Reading comprehension
- In some cases, GPT-2 outperformed **task-tuned models** — purely from pretraining and scaling.

GPT-2 Zero-shot Results

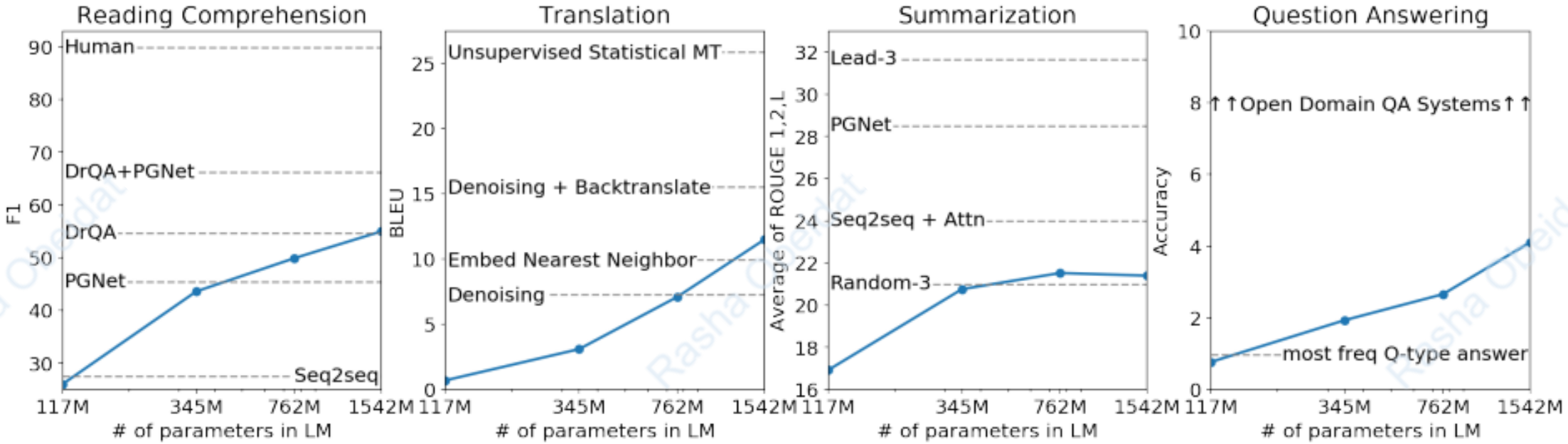


Figure 1. Zero-shot task performance of WebText LMs as a function of model size on many NLP tasks. Reading Comprehension results are on CoQA (Reddy et al., 2018), translation on WMT-14 Fr-En (Artetxe et al., 2017), summarization on CNN and Daily Mail (See et al., 2017), and Question Answering on Natural Questions (Kwiatkowski et al., 2019). Section 3 contains detailed descriptions of each result.

https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf

GPT-2 Zero-shot Results

	R-1	R-2	R-L	R-AVG
Bottom-Up Sum	**41.22**	**18.68**	**38.34**	**32.75**
Lede-3	40.38	17.66	36.62	31.55
Seq2Seq + Attn	31.33	11.81	28.83	23.99
GPT-2 `TL;DR:`	29.34	8.27	26.58	21.40
Random-3	28.78	8.63	25.52	20.98
GPT-2 no hint	21.58	4.03	19.47	15.03

Table 4. Summarization performance as measured by ROUGE F1 metrics on the CNN and Daily Mail dataset. Bottom-Up Sum is the SOTA model from (Gehrmann et al., 2018)

https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf

Moving to GPT-3 and Few-shot Learning

- Gpt-2 paper **does not formally define or evaluate** *few-shot learning.*
- *Implicitly introduce it in CoQA completion but with no analysis.*
- Few-shot prompting as a formal concept **emerged with GPT-3.**

Context (passage and previous question/answer pairs)

Tom goes everywhere with Catherine Green, a 54-year-old secretary. He moves around her office at work and goes shopping with her. "Most people don't seem to mind Tom," says Catherine, who thinks he is wonderful. "He's my fourth child," she says. She may think of him and treat him that way as her son. He moves around buying his food, paying his health bills and his taxes, but in fact Tom is a dog.

Catherine and Tom live in Sweden, a country where everyone is expected to lead an orderly life according to rules laid down by the government, which also provides a high level of care for its people. This level of care costs money.

People in Sweden pay taxes on everything, so aren't surprised to find that owning a dog means more taxes. Some people are paying as much as 500 Swedish kronor in taxes a year for the right to keep their dog, which is spent by the government on dog hospitals and sometimes medical treatment for a dog that falls ill. However, most such treatment is expensive, so owners often decide to offer health and even life _ for their dog.

In Sweden dog owners must pay for any damage their dog does. A Swedish Kennel Club official explains what this means: if your dog runs out on the road and gets hit by a passing car, you, as the owner, have to pay for any damage done to the car, even if your dog has been killed in the accident.

Q: How old is Catherine?
A: 54

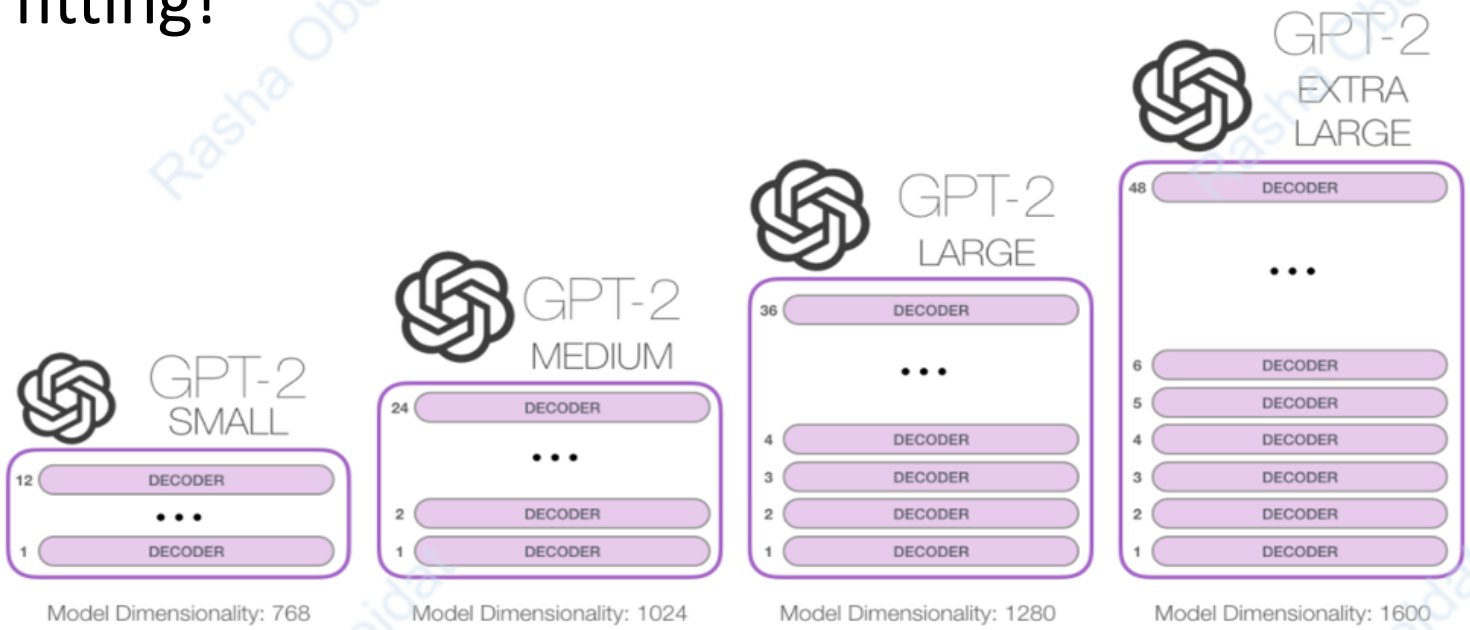
Q: where does she live?
A:

Model answer: Stockholm
Turker answers: Sweden, Sweden, in Sweden, Sweden

Table 17. Selected CoQA completion.

Moving to GPT-3 and Few-shot Learning

- Even with the GPT-2 model scaled up to 1.5 billion parameters, the WebText dataset (1.5B tokens) still wasn't large enough to fully fit the model → it is still under fitting!



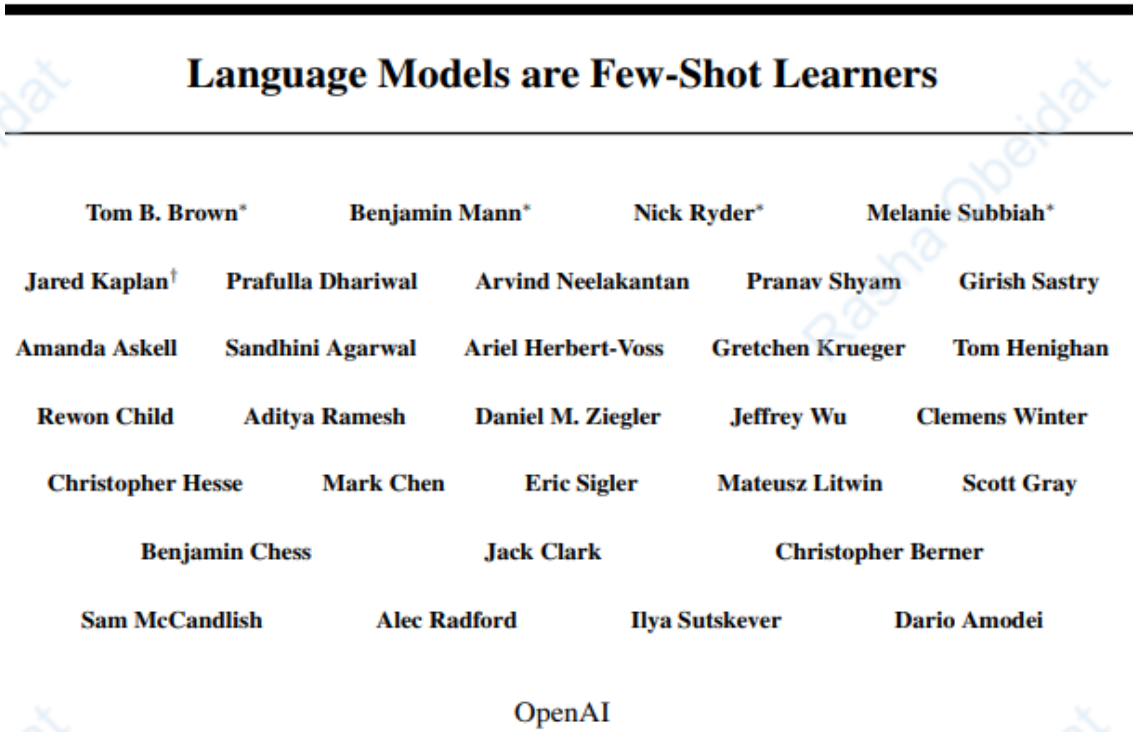
https://jalammar.github.io/illustrated-gpt2/

GPT-3

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GPT-3

- **Release:** June 2020, by OpenAI
- **Paper**: *Language Models are Few-Shot Learners.*
- **Objective**
  - Autoregressive language modeling (Same as GPT-1 and GPT-2).
  - Trained to predict the next token, using a causal (left-to-right) decoder-only transformer architecture.



Language Models are Few-Shot Learners

Tom B. Brown* Benjamin Mann* Nick Ryder* Melanie Subbiah*

Jared Kaplan† Prafulla Dhariwal Arvind Neelakantan Pranav Shyam Girish Sastry

Amanda Askell Sandhini Agarwal Ariel Herbert-Voss Gretchen Krueger Tom Henighan

Rewon Child Aditya Ramesh Daniel M. Ziegler Jeffrey Wu Clemens Winter

Christopher Hesse Mark Chen Eric Sigler Mateusz Litwin Scott Gray

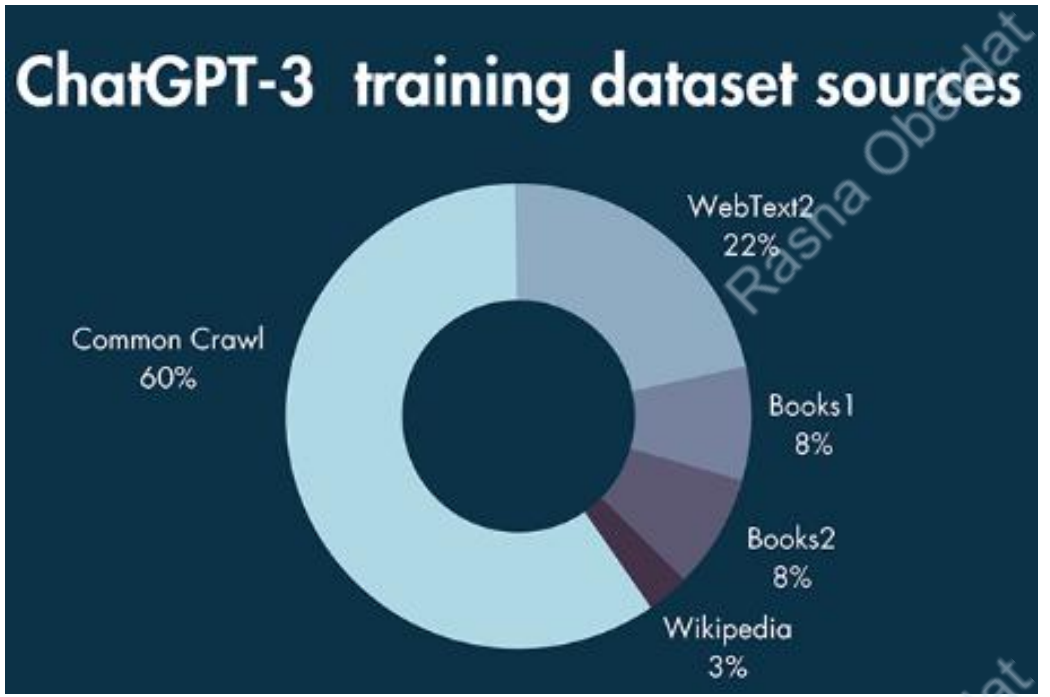
Benjamin Chess Jack Clark Christopher Berner

Sam McCandlish Alec Radford Ilya Sutskever Dario Amodei

OpenAI

GPT-3: Training Data

- **Total tokens**: **499B**
- **Mixture of datasets**:
  - Common Crawl (filtered) (~410B tokens)
  - WebText2 (an expanded version of WebText)
  - Books1 and Books2
  - Wikipedia (English)



https://www.stylefactoryproductions.com/blog/chatgpt-statistics

GPT-3 Model Scale

Version	Parameters
GPT-3 Small	125M
GPT-3 Base	350M
GPT-3 Medium	760M
GPT-3 Large	1.3B
GPT-3 XL	6.7B
GPT-3 13B	13B
GPT-3	**175B**

- **GPT-3** is a 96-layer Transformer, 12,288 hidden size, 96 attention heads (for 175B model).
- **Max sequence length:** 2048 tokens.

GPT-3 Key Idea: In-Context Learning

- The central claim of the GPT-3 paper is that **massive scale alone** enables **few-shot, one-shot, and zero-shot learning** capabilities **without any gradient updates**.
- No labeled data is required.
- **In-Context Learning**: model learns from **examples in the input prompt**, not from updating parameters
- GPT-3 behaves like a **meta-learner** by interpreting examples in context.
- **Prompting Modes**
  - **Zero-shot**: Task instruction only.
  - **One-shot**: Task instruction and *one* labeled example.
  - **Few-shot**: Task instruction and a *few* labeled example.

Examples of Prompting

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1  Translate English to French:        ← task description
2  cheese =>                           ← prompt
```

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

```
1  Translate English to French:        ← task description
2  sea otter => loutre de mer          ← example
3  cheese =>                           ← prompt
```

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
1  Translate English to French:        ← task description
2  sea otter => loutre de mer          ← examples
3  peppermint => menthe poivrée        ←
4  plush girafe => girafe peluche      ←
5  cheese =>                           ← prompt
```

https://arxiv.org/pdf/2005.14165

What changed from GPT-2 to GPT-3?

- GPT-2 and GPT-3 have the:
  - Same architecture (decoder-only transformer).
  - Same objective (next-token prediction).
  - No fine-tuning.
- But two major things are new:

  1. Massive Scale in number of parameters and pretraining dataset.

     This allowed the model to memorize many task patterns implicitly during pretraining.

  2. Formal Introduction of **In-Context** Learning.

  - GPT-3 was the first to show that a model trained on just next-token prediction could perform a wide range of unseen tasks if examples are provided in the prompt (few-shot) even with no parameter updates.
  - This behavior was not shown or evaluated in GPT-2 systematically.

Why is it called "In-Context Learning"?

Because the model:

- Doesn't learn by updating weights (no SGD/fine-tuning).
- Instead, it learns from examples presented in the prompt, *within the **context window**.*
- The model "adapts" to the task **within the inference context**, not during training.
- It's like meta-learning without gradient updates.

GPT-3: Performance Comparison Results

- In the original GPT-3 paper:
  - **In-context learning** (zero-, one-, and few-shot prompting) achieved SOTA or near-SOTA performance on:
    - TriviaQA (zero-shot QA).
    - LAMBADA (text completion).
    - Translation benchmarks (few-shot).
    - Reading comprehension.
  - Few-shot prompting (2-32 examples in the prompt) generally performs **much better** than zero-shot.

GPT-3: Performance on NLU tasks

	SuperGLUE Average	BoolQ Accuracy	CB Accuracy	CB F1	COPA Accuracy	RTE Accuracy
Fine-tuned SOTA	**89.0**	**91.0**	**96.9**	**93.9**	**94.8**	**92.5**
Fine-tuned BERT-Large	69.0	77.4	83.6	75.7	70.6	71.7
GPT-3 Few-Shot	71.8	76.4	75.6	52.0	92.0	69.0

	WiC Accuracy	WSC Accuracy	MultiRC Accuracy	MultiRC F1a	ReCoRD Accuracy	ReCoRD F1
Fine-tuned SOTA	**76.1**	**93.8**	**62.3**	**88.2**	**92.5**	**93.3**
Fine-tuned BERT-Large	69.6	64.6	24.1	70.0	71.3	72.0
GPT-3 Few-Shot	49.4	80.1	30.5	75.4	90.2	91.1

Table 3.8: Performance of GPT-3 on SuperGLUE compared to fine-tuned baselines and SOTA. All results are reported on the test set. GPT-3 few-shot is given a total of 32 examples within the context of each task and performs no gradient updates.

GPT-3: Performance on QA tasks

Setting	PIQA	ARC (Easy)	ARC (Challenge)	OpenBookQA
Fine-tuned SOTA	79.4	**92.0**[KKS+20]	**78.5**[KKS+20]	**87.2**[KKS+20]
GPT-3 Zero-Shot	**80.5***	68.8	51.4	57.6
GPT-3 One-Shot	**80.5***	71.2	53.2	58.8
GPT-3 Few-Shot	**82.8***	70.1	51.5	65.4

Table 3.6: GPT-3 results on three commonsense reasoning tasks, PIQA, ARC, and OpenBookQA. GPT-3 Few-Shot PIQA result is evaluated on the test server. See Section 4 for details on potential contamination issues on the PIQA test set.

GPT-3 Limitations

Limitations Noted in Paper

1. Performance still lags behind fine-tuned models on some tasks.

2. Very **sensitive to prompt wording** and ordering.

GPT-4

- **Release**: March 2023 (OpenAI).
- **Architecture**: Multimodal (can process text + images).
- **Model Type**: Transformer-based, decoder-only (details not fully. disclosed)
- **Training Objective**: Same causal next-token prediction.
- **Token Limit**: Up to 32,000 tokens (for pro users).

Model	Max Context Window (Token Limit)
GPT-1	512 tokens
GPT-2	1,024 tokens
GPT-3	2,048 tokens
GPT-4	Up to 32,000 tokens (depending on version)

Dr. Rasha Obeidat ®

GPT-4

- **Performance**:
  - Stronger than GPT-3 on reasoning, coding, and multilingual tasks
  - Better alignment via ***reinforcement learning*** and ***preference tuning*** (used in ChatGPT).
- **Capabilities**:
  - Handles more nuanced instructions.
  - More reliable at following task constraints.
  - Can solve complex problems with greater accuracy.

Dr. Rasha Obeidat ®

InstructGPT

- **InstructGPT** is a fine-tuned version of GPT-3 trained to follow human instructions more accurately.
- Introduced by OpenAI in early 2022.
- Training process involves:
  - **Step 1**: Supervised fine-tuning on instruction-following examples.
  - **Step 2**: Reinforcement Learning from Human Feedback (**RLHF**).
    - Human labelers rank model outputs.
    - A reward model is trained from this ranking.
    - GPT-3 is fine-tuned using Proximal Policy Optimization (PPO).
- InstructGPT is the basis for **ChatGPT**, showing dramatic improvements over plain GPT-3 in following prompts.

Supervised Instruction Tuning

- Supervised Instruction Tuning is the **first stage** in training InstructGPT.
- A dataset is curated with **instruction-following examples**, created by human annotators.
- The model (e.g., GPT-3) is then **fine-tuned using supervised learning**, where:
  - **Input**: **A task Instruction + Input Example** (as a single prompt).
  - **Target**: The ideal human-written response.
- The model learns to **minimize the difference (loss)** between its output and the target response.

Example:

Prompt:
"Translate the following sentence to Spanish:
'The meeting starts at noon.'"

Target Output:
"سيبدأ الاجتماع قريبا."

- The model is trained to produce this output directly, learning the behavior of instruction-following.

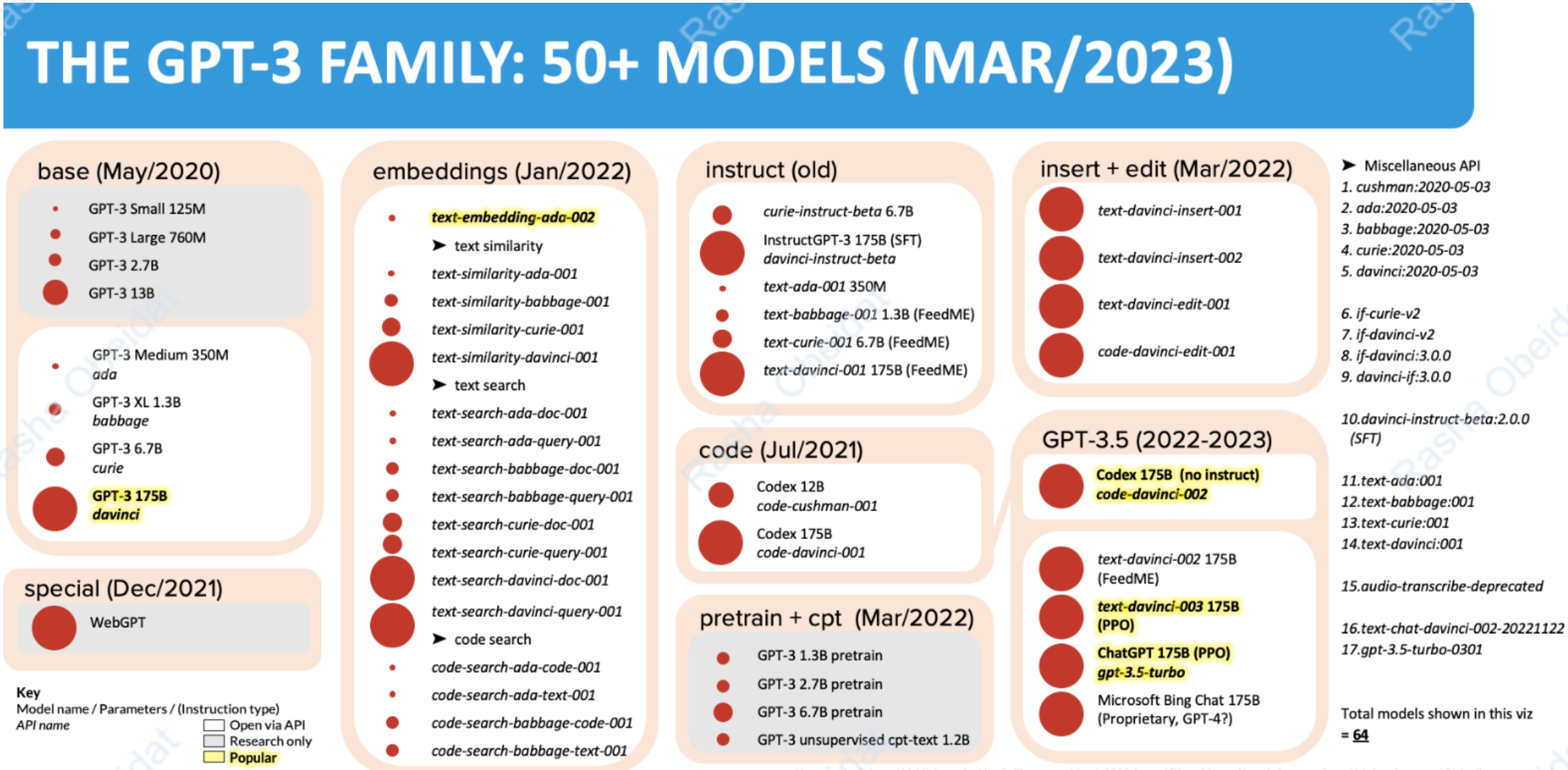
Chat-GPT

- **ChatGPT** is a conversational AI interface based on models like GPT-3.5 or GPT-4.
- While **GPT-3/4** are raw pretrained models, ChatGPT is **instruction-tuned** to follow human prompts more helpfully.
- It builds on **InstructGPT**, which was fine-tuned from GPT-3 using **supervised Instruction tuning** followed by **reinforcement learning from human feedback (RLHF).**
- ChatGPT differs in:
  - **Tone**: More aligned with user intent.
  - **Behavior**: more conversational.
  - **Interface**: Deployed as a chatbot, maintains dialogue context.
- In essence, ChatGPT = GPT-3.5/4 + instruction tuning + RLHF.

RLHF

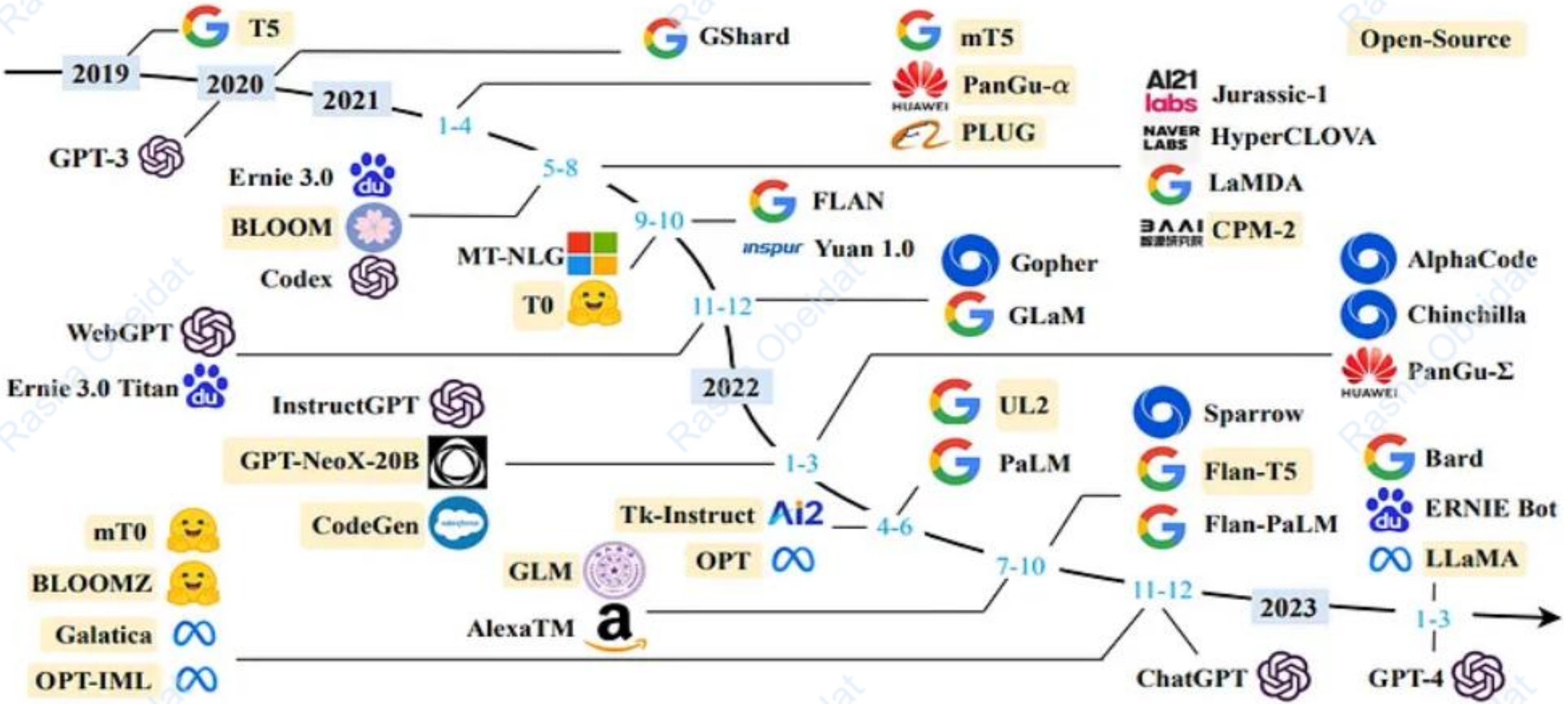
- **RLHF stands for** Reinforcement Learning from Human Feedback.
- RLHF is the second stage of InstructGPT training, applied after supervised fine-tuning.
- **Goal**: Align model behavior with human preferences (helpfulness, harmlessness, honesty).
- **Three Steps:**
  1. **Collect human preferences**:
     1. Show annotators multiple model completions.
     2. Ask them to **rank** outputs from best to worst.
  2. **Train a reward model**:
     1. Use rankings to train a model that predicts human preferences.
  3. **Fine-tune the model**:
     1. Use **Proximal Policy Optimization (PPO)** to train GPT with the reward model as a signal.
- **Example:**
  - Prompt: "Explain the water cycle“.
  - Model generates 3 answers.
  - Human ranks them → Reward model learns this ranking → Model is fine-tuned to produce better completions.

OpenAI LLMs landscape 2023



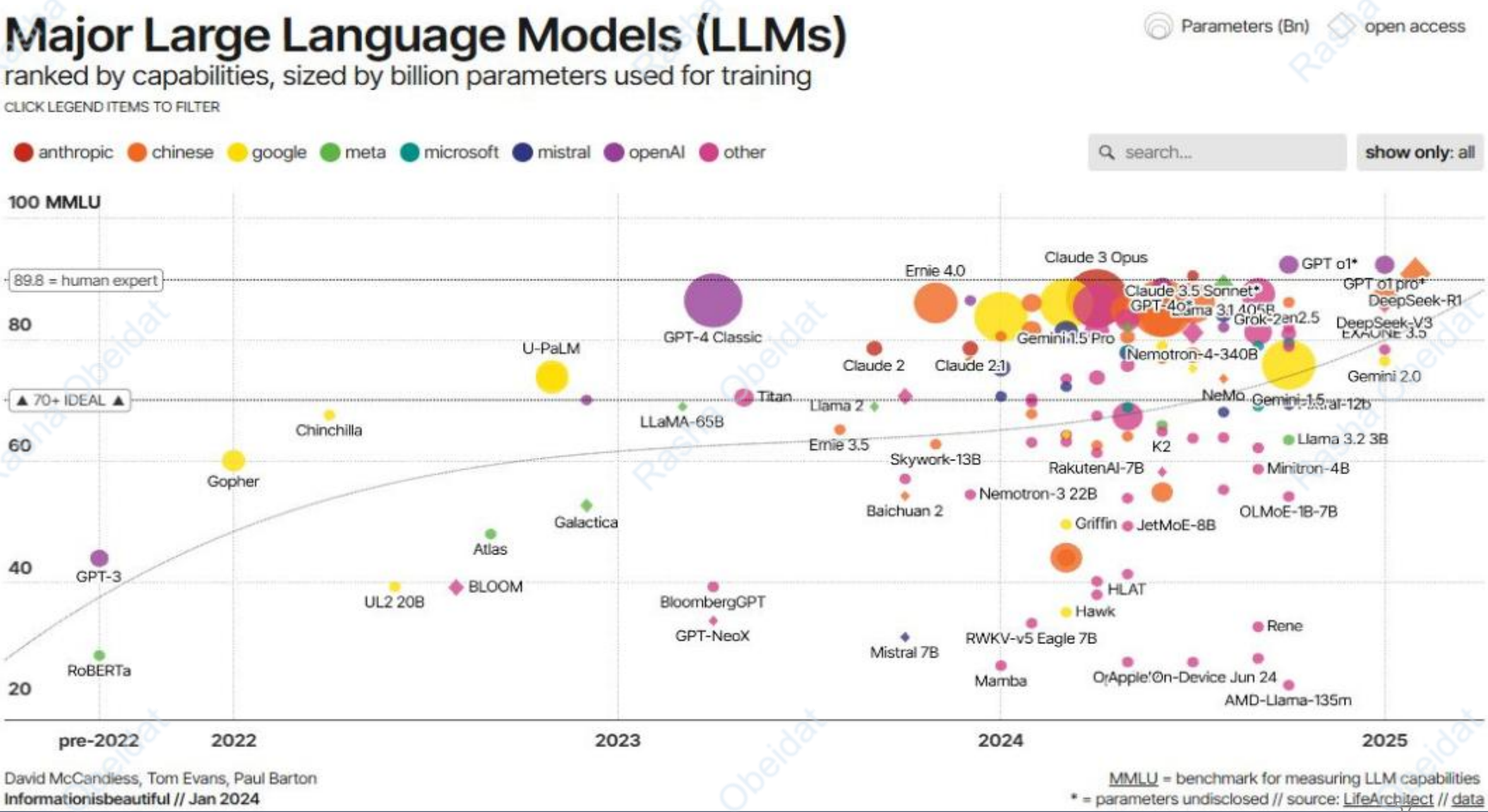
https://s10251.pcdn.co/wp-content/uploads/2023/03/2023-Alan-D-Thompson-GPT3-Family-Rev-1.png

LLMs landscape 2023



https://www.linkedin.com/pulse/future-large-language-models-evolution-specialization-shekhar-meena/

LLMs landscape 2023



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